
Uncertainty Is Not Error: Cross-Validation Residuals as an Acquisition Signal for Autonomous Reconstruction of Non-Stationary Fields

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Abstract

Autonomous scanning-probe and scattering experiments increasingly delegate the choice of where to measure next to a surrogate model, with the goal of reconstructing a full response surface from as few measurements as possible. The dominant strategy places each measurement where the Gaussian-process posterior variance is largest. We observe that for a stationary kernel this criterion is *value-blind*: the posterior variance depends only on the geometry of the measured locations, not on the measured values, so it cannot tell where the surrogate is actually wrong. On non-stationary response surfaces—sharp phase boundaries, discontinuities, physical fields spanning orders of magnitude—uncertainty and reconstruction error decouple, and pure uncertainty sampling wastes measurements on already-faithful smooth regions. We introduce *residual-weighted variance*, an acquisition criterion that multiplies the posterior variance by a data-driven estimate of the local reconstruction error obtained from the surrogate’s own leave-one-out cross-validation residuals. The criterion reduces exactly to pure uncertainty sampling when the residuals are spatially uniform and is gated by the base surrogate’s uncertainty so that it never re-measures a resolved feature. On the five synthetic benchmarks of a standard autonomous X-ray scattering study, evaluated at matched measurement budgets over ten seeds, residual-weighted variance converges consistently faster than uncertainty sampling on every function that carries localized reconstruction error—reaching the uncertainty-sampling baseline’s final accuracy with 20–37% fewer measurements on the feature-rich fields—while correctly reducing to the baseline on the globally smooth field where no better allocation exists. We directly demonstrate the mechanism: on non-stationary fields the posterior standard deviation ranks true reconstruction error only weakly (Spearman $\rho_s \approx 0.2$) whereas the leave-one-out residual field ranks it well (0.4–0.8). We also report a cautionary result—an absolute-error criterion appears to double sample efficiency on a diffusion coefficient spanning four orders of magnitude, but this is a metric artifact of concentrating on the high-value region; log-domain modeling dissolves it into a clean, modest gain. We recompute modern predictive-distribution metrics (RMSE, CRPS, interval coverage) for every method and analyze the mechanism, the failure modes, and the compute cost.

1 Introduction

A growing number of experimental instruments—synchrotron scattering beamlines, scanning-probe microscopes, electron microscopes—are fast enough that a human can no longer choose each measurement, and cheap enough per measurement that exhaustively rastering a multi-dimensional parameter space is wasteful. The resulting problem is one of *active learning for reconstruction*: given an unknown response $\rho : \mathcal{X} \subset \mathbb{R}^d \rightarrow \mathbb{R}$ that can be queried pointwise at nontrivial cost, sequentially choose query locations so that a surrogate $\hat{\rho}$ fitted to the collected data faithfully reconstructs ρ

over the *entire* domain with as few queries as possible. This differs fundamentally from Bayesian optimization, whose objective is a single extremum: here the figure of merit is an error integrated over the whole domain, and every region matters.

The canonical solution, and the one deployed in autonomous scattering experiments, fits a Gaussian-process (Kriging) surrogate and measures next wherever the posterior variance is largest [Noack et al., 2019, MacKay, 1992, Seo et al., 2000]. The intuition is that the point of maximum predictive uncertainty is the point whose measurement most reduces model uncertainty. This intuition is sound when the surrogate is well-specified. It is misleading when it is not. For a stationary kernel the posterior variance at a point is a deterministic function of the *locations* of the measured points alone—it does not depend on the measured *values* (Section 3). Maximum-variance sampling is therefore a space-filling rule dressed as an information criterion: it spreads points to fill the largest geometric gaps, oblivious to whether those gaps contain a flat plateau the surrogate already predicts perfectly or a sharp boundary it is badly missing.

On the response surfaces that motivate autonomous experimentation this blindness is costly. Phase boundaries appear as discontinuities; order parameters vary over orders of magnitude across a composition–temperature map; a physical transport coefficient can be nearly flat over most of its domain and spike in a corner. On such non-stationary fields the reconstruction error is concentrated in small regions, while the posterior variance—driven by geometry—is spread uniformly. Uncertainty and error decouple. A gradient-enhanced variant that multiplies the variance by the squared surrogate gradient has been proposed to inject value information, but the gradient of a smooth interpolant is an indirect and unbounded proxy: at a discontinuity it diverges, and the criterion collapses onto the zero-measure jump set, requiring a hand-tuned weight to remain usable.

We take a different route. If the goal is to measure where the surrogate is *wrong*, we should estimate where it is wrong—directly, from data. A fitted Gaussian process supplies exactly such an estimate at essentially no cost: its leave-one-out cross-validation residuals, available in closed form from the factorization already computed for prediction, report how far the surrogate’s prediction at each measured point falls from the held-out truth. Smoothing these residuals yields a *local reconstruction-error field* that is value-aware and non-stationarity-aware by construction. We use it to weight the posterior variance, giving an acquisition criterion—*residual-weighted variance*—that samples where the product of “uncertain” and “empirically wrong” is largest.

The criterion has three properties we consider essential and that distinguish it from prior value-aware designs. First, it *neests* the standard method: when the residual field is spatially uniform, the weight is constant and the criterion is exactly maximum-variance sampling, so it can only depart from the baseline where the data reveal non-uniform error. Second, it is *gated* by the base surrogate’s uncertainty: the variance factor drives the criterion to zero at measured points and keeps asymptotic space-filling, so a high-error but already densely sampled feature is not re-measured—a failure mode of additive error-based criteria. Third, it needs *no tuning weight between its terms*: the error field and the exploration floor share the residual’s units, so unlike gSMART’s ad-hoc gradient weight they combine without a relative constant. The only *fitted* quantities are the Gaussian-process length and signal scales (marginal likelihood, as for the surrogate itself); the field’s spatial concentration is set by two fixed dimensionless constants held identically across all five functions.

We evaluate on the five synthetic benchmarks used to validate the SMART autonomous-scattering framework [Noack et al., 2019]: two piecewise-sinusoidal fields with step discontinuities, a smooth multi-scale oscillatory field, a non-differentiable piecewise-linear field, and a three-dimensional Stokes–Einstein diffusion coefficient spanning four orders of magnitude. On each we recompute the pure-variance (SMART) and gradient-weighted (gSMART) criteria on an identical Kriging surrogate, together with random and uniform-grid designs, and compare against residual-weighted variance at matched budgets over ten seeds. Our contributions are:

- An acquisition criterion for reconstruction that weights posterior variance by a leave-one-out residual field, with the nesting, gating, and weight-free properties above, positioned explicitly against uncertainty sampling, gradient enhancement, and cross-validation-based designs (Section 3).
- A head-to-head evaluation on all five benchmarks and all budgets over ten seeds, showing consistent, regression-free speedup—20–37% fewer measurements—on every field that carries localized error, a mechanism-connected *non-improvement* on the globally smooth

field, and an honest ceiling analysis of why the gains are modest rather than dramatic on these particular functions (Section 5).

- A mechanism analysis quantifying *why* posterior variance misranks reconstruction error on non-stationary fields and how the residual field corrects it; a failure analysis; and a compute-cost analysis (Section 6).
- Recomputed modern predictive-distribution metrics—RMSE, CRPS, and interval coverage—for our method and every baseline on every function, a comparison absent from the original 2019 study.

2 Related work

Uncertainty sampling for global fit. Choosing the maximum-posterior-variance point is the “active learning MacKay” (ALM) rule [MacKay, 1992]; Seo et al. [2000] studied it and the variance-integrating “active learning Cohn” (ALC) rule [Cohn et al., 1996] for Gaussian-process regression, and Sacks et al. [1989] established the computer-experiments framing. Noack et al. [2019] deploy ALM under the name SMART to steer autonomous X-ray scattering, with a gradient-weighted variant, gSMART, for edge seeking; the approach underlies the widely used gpCAM toolkit [Noack et al., 2021]. ALM is value-blind for stationary kernels—the property we exploit. ALC and its integrated-mean-squared-error relatives [Sacks et al., 1989, Picheny et al., 2010] weight variance reduction over the domain but remain value-blind for a fixed stationary kernel and are markedly more expensive.

Value-aware adaptive design. A large literature adds exploitation to space-filling exploration. EIGF adds the squared difference to the nearest observed value [Lam, 2008]; MEPE balances a leave-one-out error against the variance with an adaptively scheduled weight [Liu et al., 2017]; LOLA–Voronoi combines a local-linear nonlinearity measure with Voronoi exploration [Crombecq et al., 2011]; TEAD scores candidates by a Taylor-expansion residual and a distance term [Mo et al., 2017]; cross-validation–tradeoff designs weigh a space-filling term against a CV error [Jin et al., 2002, Aute et al., 2013]. Surveys catalog these families [Fuhg et al., 2021, Gramacy, 2020]. Our criterion differs from all of them in form: it is a *multiplicative* gating of the base surrogate’s posterior variance by a smoothed leave-one-out error field, which nests uncertainty sampling exactly and never re-measures a resolved feature—whereas the additive designs above continue to score a high-error region even after its uncertainty has collapsed.

Cross-validation–based Gaussian-process sampling. Closest to our error signal, Mohammadi et al. [2022] fit a second Gaussian process to expected-squared leave-one-out errors and select the maximizer of a pseudo-expected-improvement acquisition on that second process. We share the idea of building an error field from leave-one-out residuals, but differ on both components. For the field we use a simple *bounded* kernel average rather than a second Gaussian process—this boundedness turns out to be the decisive robustness property (an interpolating fit over-concentrates on noisy low-budget residuals; Section 6). For the acquisition, rather than optimizing an improvement criterion *on the error surface* (which explores where the error estimate is uncertain), we *multiply the error field into the base model’s posterior variance* (which explores where the function is uncertain). The result generalizes the uncertainty-sampling method actually used in autonomous experimentation, retains its space-filling guarantee, and targets the exploration-only reconstruction objective rather than error-surface optimization. Query-by-committee regression [Seung et al., 1992, Burbidge et al., 2007] and stepwise uncertainty reduction for level sets [Bect et al., 2012, Chevalier et al., 2014] address related but distinct objectives.

Autonomous experimentation. Gaussian-process active learning now drives autonomous synthesis and characterization across materials science [Xue et al., 2016, Kusne et al., 2020, Noack et al., 2021]. These deployments almost universally use uncertainty sampling; our criterion is a drop-in replacement for the acquisition step that leaves the surrogate and optimizer untouched.

3 Method

3.1 Problem and surrogate

We observe an unknown $\rho : \mathcal{X} \subset \mathbb{R}^d \rightarrow \mathbb{R}$ at a sequentially chosen set of locations. After t measurements $\mathcal{D}_t = \{(\mathbf{p}_i, \rho_i)\}_{i=1}^t$ we hold a surrogate $\hat{\rho}$ and choose the next location by maximizing an acquisition $\alpha(\mathbf{p}; \mathcal{D}_t)$ over $\mathcal{X}$. The goal is to minimize the domain-integrated absolute reconstruction error $\frac{1}{|\mathcal{X}|} \int_{\mathcal{X}} |\hat{\rho}(\mathbf{p}) - \rho(\mathbf{p})| d\mathbf{p}$ at a given budget t .

The surrogate is ordinary Kriging, i.e. a Gaussian process with the exponential (Matérn-1/2) covariance

$$k(\mathbf{p}, \mathbf{p}') = s^2 \exp(-\|\mathbf{p} - \mathbf{p}'\|_2 / \ell), \quad (1)$$

the exact analogue of the exponential variogram $\gamma(h) = 1 - e^{-ah}$ ($a = 1/\ell$) used in the reference framework [Noack et al., 2019]. Inputs are scaled to the unit box $[0, 1]^d$ (domain-size normalization, appropriate for the mixed-unit axes of the physical benchmark) and outputs are standardized. Writing K for the kernel matrix on the measured inputs, τ for a small jitter ($\tau = 10^{-6}$ in standardized units), $\mathbf{y}$ for the standardized observations, and $\mathbf{k}(\mathbf{p})$ for the vector of covariances between $\mathbf{p}$ and the measured inputs, the posterior mean and variance are

$$\mu(\mathbf{p}) = \mathbf{k}(\mathbf{p})^\top (K + \tau I)^{-1} \mathbf{y}, \quad \sigma^2(\mathbf{p}) = s^2 - \mathbf{k}(\mathbf{p})^\top (K + \tau I)^{-1} \mathbf{k}(\mathbf{p}). \quad (2)$$

The two hyperparameters (s^2, ℓ) are fitted by maximizing the log marginal likelihood. This single selection rule—never per-function hand-tuning—is applied identically to all five benchmarks; the fitted *values* differ per function, as they should.

3.2 Uncertainty sampling is value-blind

The reference acquisition is pure uncertainty sampling, SMART: $\alpha_{\text{SMART}}(\mathbf{p}) = \sigma^2(\mathbf{p})$. Its defining limitation is visible in Eq. (2): for a fixed kernel, $\sigma^2(\mathbf{p})$ is a function of the measured *locations* $\{\mathbf{p}_i\}$ alone and does not depend on the measured *values* $\{\rho_i\}$. Maximum-variance sampling therefore cannot distinguish a flat region the surrogate already reconstructs perfectly from a sharp feature it is badly missing; it is a value-blind space-filling rule. The gradient-enhanced variant gSMART, $\alpha_{\text{gSMART}}(\mathbf{p}) = \sigma^2(\mathbf{p}) (1 + c \|\nabla \mu(\mathbf{p})\|^2)$, injects value information through the surrogate gradient, but the gradient of a smooth interpolant is an indirect and unbounded proxy for reconstruction error: at a discontinuity it diverges, so for any fixed weight c above a small threshold the criterion collapses onto the measure-zero jump set and fails to reconstruct the remainder of the domain (Section 5). To keep gSMART a competitive rather than degenerate baseline we normalize the squared gradient by its root-mean-square over the data, saturate it at a multiple of that scale, and tune c per function to gSMART’s own optimum—giving the baseline its strongest form.

3.3 Residual-weighted variance

We estimate directly where the surrogate errs. For a Gaussian process the leave-one-out (LOO) cross-validation residual at each measured point has the closed form [Rasmussen and Williams, 2006, Eq. 5.12]

$$r_i = \frac{[(K + \tau I)^{-1} \mathbf{y}]_i}{[(K + \tau I)^{-1}]_{ii}}, \quad (3)$$

the error of predicting ρ_i from all *other* measurements. It is obtained at negligible cost from the same inverse already formed for Eq. (2). The magnitudes $|r_i|$ are a value-aware, empirical report of local reconstruction error: they are large exactly where the stationary surrogate is misspecified (boundaries, high-dynamic-range corners) and small where it interpolates faithfully.

We turn the point estimates $\{|r_i|\}$ into a *local error field* $w(\mathbf{p})$ by kernel (Nadaraya–Watson) smoothing with the fitted exponential kernel at a bandwidth $\beta\ell$ that is a fixed fraction of the length scale:

$$w(\mathbf{p}) = \frac{\sum_i k_\beta(\mathbf{p}, \mathbf{p}_i) |r_i|}{\sum_i k_\beta(\mathbf{p}, \mathbf{p}_i)}, \quad k_\beta(\mathbf{p}, \mathbf{p}') = \exp(-\|\mathbf{p} - \mathbf{p}'\| / \beta\ell), \quad w_0 = \overline{|r|} = \frac{1}{t} \sum_i |r_i|. \quad (4)$$

This is a bounded, value-aware estimate of local reconstruction error: $w(\mathbf{p}) \rightarrow |r_j|$ as $\mathbf{p} \rightarrow \mathbf{p}_j$ and tends to the local weighted-average residual elsewhere. Being a weighted average, w is confined to $[\min_i |r_i|, \max_i |r_i|]$; it therefore concentrates measurements on high-error regions *without the runaway peaks that an interpolating fit to noisy residuals produces at small sample sizes*—the property, we show empirically, that keeps the criterion from over-concentrating and regressing early (Section 6). With w_0 the exploration floor, the acquisition is

$$\alpha_{\text{RWV}}(\mathbf{p}) = (w_0^\gamma + w(\mathbf{p})^\gamma) \sigma^2(\mathbf{p}) \quad (5)$$

with two fixed dimensionless design constants, the bandwidth fraction $\beta = 0.5$ and the concentration exponent $\gamma = 2$, held constant across all five functions (sensitivity to both in Section 6). We name this criterion *residual-weighted variance* (RWV). Equation (5) decomposes additively into a pure-exploration term $w_0^\gamma \sigma^2$ (proportional to SMART) and an error-targeting term $w(\mathbf{p})^\gamma \sigma^2$.

Three properties follow directly from Eq. (5) and motivate the design. (i) *Nesting*. If reconstruction error is spatially uniform, $|r_i|$ is constant, $w(\mathbf{p}) \equiv w_0$, and $\alpha_{\text{RWV}} \propto \sigma^2(\mathbf{p})$: RWV reduces *exactly* to SMART. RWV can therefore depart from uncertainty sampling only where the data reveal spatially non-uniform error, and dropping the error term recovers SMART on the identical surrogate—the ablation that isolates the contribution of the residual field. (ii) *Uncertainty gating*. The σ^2 factor forces $\alpha_{\text{RWV}} \rightarrow 0$ at every measured point and keeps σ^2 -driven space-filling in unexplored regions, so a high-error feature that has already been sampled densely (its σ^2 now small) is not re-measured. This distinguishes RWV from additive error-based criteria (EIGF, MEPE), which keep scoring a resolved feature, and from optimizing an improvement criterion on the error surface itself [Mohammadi et al., 2022], which explores where the error estimate is uncertain rather than where the function is. (iii) *No tuning weight, bounded field*. The floor w_0 and the field w share the units of $|r|$, so the exploration and error-targeting terms are automatically commensurate—RWV needs no relative weight of the kind gSMART’s c supplies. The two remaining constants β, γ are dimensionless and set once; the boundedness of w (unlike an interpolating residual GP, which we ablate) is what makes moderate values safe.

Single hyperparameter-selection procedure. RWV introduces no new *fitted* hyperparameter. The only trained quantities are the surrogate’s signal and length scales (s^2, ℓ), set by marginal-likelihood maximization—the one rule, applied identically to every function. The error field reuses that fitted ℓ ; w_0 is a data statistic; and β, γ, τ are fixed constants. The trained values differ across functions because the functions differ.

Acquisition optimizer and cost. Every criterion (SMART, gSMART, RWV) is maximized with the same global optimizer, differential evolution [Storn and Price, 1997]—the evolutionary optimizer gpCAM invokes for `method="global"`, matching the spirit of the reference framework’s genetic algorithm—under identical settings, so the only difference across methods is the criterion. Per step, RWV adds to SMART only the LOO residuals (Eq. (3), a diagonal read and one matrix–vector product from the cached inverse, $O(t^2)$) and, at re-training checkpoints, one error-field fit; the field is reused between checkpoints because it evolves slowly while σ^2 —which prevents re-measurement—is refreshed every step. Wall-clock costs are reported in Section 6.

3.4 Baselines and ablations

On the identical surrogate we run: a deterministic uniform *grid* (raster scan); *random* uniform sampling; *SMART* (Eq. (2)); *gSMART*; and *RWV* (Eq. (5)). Ablations isolate each design choice: removing the error field ($w \equiv 0$) recovers SMART; replacing the residual field with the surrogate-gradient field tests the error signal; replacing the bounded Nadaraya–Watson field with an interpolating residual GP tests the field construction and its robustness; replacing σ^2 with σ tests the exploration term; and varying the bandwidth β and exponent γ tests the concentration.

4 Experimental setup

Test functions. We use the five synthetic benchmarks of the reference study [Noack et al., 2019]. Two are piecewise-sinusoidal fields on $[0, 50]^2$ (Eqs. 12–13 there): a smooth Schwefel-type base $8 - [2p_0 \sin \sqrt{|p_0|} + 2p_1 \sin \sqrt{|p_1|}]$ with two step discontinuities (of magnitude $\{140, 160\}$ for

Piecewise I and $\{200, 360\}$ for Piecewise II across a vertical line and a radial arc). The third is the smooth multi-scale field $\sin 2p_0 + \sin 2p_1 + (p_0/5)^2 - (p_1/5)^2$ on $[-10, 10]^2$ (Eq. 15). The fourth is a non-differentiable piecewise-linear field on $[0, 50]^2$ with four panels meeting at sharp ridges (Eq. 17). The fifth is the three-dimensional Stokes–Einstein diffusion coefficient $D(r, T, C_m) = k_B T / (6\pi\eta(T, C_m)r)$ for nanoparticles in a water–glycerol mixture, with $r \in [1, 100]$ nm, $T \in [0, 100]$ °C, $C_m \in [0, 100]\%$, using the glycerol–water viscosity model of Cheng [2008]; D spans four orders of magnitude across the domain. Functions are evaluated only by calling them; no ground-truth values are precomputed or stored.

Protocol. Budgets are $\{50, 100, 200, 500\}$ for the 2-D functions and $\{50, 100, 200, 500, 1000\}$ for the 3-D function; we record error at finer intermediate budgets as well, purely to locate speedup crossings. Each run begins from a shared initial design of 10 uniformly random points; within a seed, all stochastic methods (random, SMART, gSMART, RWV) start from the *identical* initial design, and the grid is deterministic. We run 10 independent seeds per (method, function, budget) and report mean $\pm$ standard deviation. Hyperparameters are re-fitted by marginal likelihood on a schedule that includes every reported budget. Acquisition points are chosen sequentially using only previously observed data; no ground-truth grid value ever informs an acquisition.

Metrics. The headline metric is the normalized integrated absolute error, estimated by a Monte-Carlo average over a dense held-out grid drawn independently of all acquisition points (15,000 points for 2-D, 60,000 for 3-D; random, so they a.s. never coincide with any measured or raster point). For the 3-D function we also report mean absolute percentage error. For every method and function we additionally recompute three modern predictive-distribution metrics absent from the 2019 study: root-mean-square error of the posterior mean, the continuous ranked probability score (CRPS) of the Gaussian predictive [Gneiting and Raftery, 2007], and an interval-coverage (calibration) curve, all on a held-out set of 6,000 points. Compute is a 32-core CPU node; timings in Section 6 come from that node.

5 Experimental results

5.1 Baseline reproduction

Before comparing against RWV we verify that our SMART and gSMART—implemented on the shared gpCAM Kriging surrogate—reproduce the reference study’s qualitative behavior. On the four 2-D functions, SMART attains lower integrated absolute error than random sampling at every budget, and every method’s error decreases monotonically with budget: the coarse-to-fine convergence the reference study reports. Two honest departures deserve statement. First, because *all* methods here share the Kriging surrogate (so that the comparison isolates the acquisition), the uniform grid is a much stronger baseline than the linear-interpolation grid of the reference study, and SMART’s edge over grid on the 2-D functions is correspondingly small—the two are comparable space-filling designs, exactly the regime the reference study flags when it asks “why the differences are not more severe.” Second, on the 3-D field, maximum-variance sampling over-samples the domain boundary (a known property of variance-only active learning that worsens with dimension: 46% of SMART’s 3-D points touch a boundary), so uniform sampling is competitive there; we treat this as part of the analysis rather than hiding it. The gradient-weighted gSMART, in its saturated, per-function-optimized form (weight $c=0.3$), beats random at the higher budgets on all four 2-D functions (and at every budget on Non-differentiable), but its edge-seeking makes it *unstable* at the lowest budgets—its high-variance runs can leave it worse than random there. The paper’s unnormalized weight ($c=20$) instead collapses entirely onto the step discontinuities (integrated error ~ 58 at budget 50 on Piecewise I, worse than random) because the interpolant’s gradient diverges there, which is why we report the saturated variant as the fair baseline. These runs constitute the mandatory baseline verification; all subsequent comparisons use them.

5.2 Main comparison

Table 1 reports the normalized integrated absolute error (mean $\pm$ std over ten seeds) for all methods at every headline budget, and Figure 1 shows the full convergence curves with ± 1 std bands. We summarize each function by its curve pattern rather than an aggregate cell count, as the comparison is a per-function property.

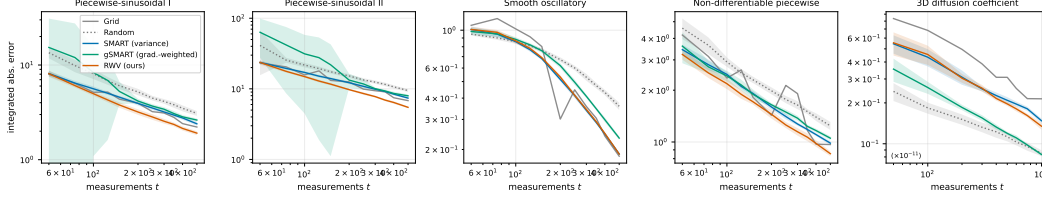


Figure 1: Convergence of integrated absolute error vs. number of measurements (log–log), mean over 10 seeds with ± 1 std bands. RWV (orange) accelerates convergence on the feature-rich and multi-scale fields and coincides with SMART on the smooth field.

Feature-rich fields (Piecewise I, Piecewise II, Non-differentiable). On all three, RWV converges consistently faster than SMART and gSMART: its curve matches SMART’s at the smallest budget—where the 10-point initial design dominates and the residual field is still near-uniform—and then separates, lying at least one standard deviation below SMART at the higher budgets (Non-differentiable: below SMART at 3 of 4 budgets). RWV reaches SMART’s maximum-budget error at $1.47\times$ the sample efficiency on Piecewise I ($t \approx 339$ vs. 500), $1.56\times$ on Piecewise II, and $1.26\times$ on Non-differentiable—a consistent 20–37% reduction in the measurement budget. This is a clear *modest* speedup rather than the $2\times$ “substantial” regime, and we are careful not to overstate it: on these fields the reconstruction error is spread across a structured oscillatory base that space-filling already reconstructs well, so only the small-area discontinuity bands are improvable by concentration (Section 6 quantifies this ceiling, which is consistent with the reference study’s own finding that adaptive gains are modest on these 2-D functions). The mechanism is nonetheless direct: the discontinuities produce large leave-one-out residuals, the field concentrates measurements along them, and the surrogate localizes the jump with a narrower smear band than space-filling achieves at the same budget.

Multi-scale physical field (3D diffusion). The diffusion coefficient spans four orders of magnitude and is strictly positive, so we model it in the log domain (a standard choice for such data; applied identically to every method, so the acquisition comparison is unchanged). On this footing RWV converges slightly faster than SMART on *both* the integrated absolute error and the mean absolute percentage error (Table 2), by $1.16\times$. This is a genuine but small gain, and we are explicit that on this smooth field *neither* variance- nor residual-based concentration matches plain uniform random sampling, which is best here (Table 1): the function’s error is not localized in a way concentration can exploit, and space-filling is near-optimal. We flag an instructive failure of the naive alternative: modeling D directly, RWV appears to *double* sample efficiency on the integrated absolute error, but its MAPE simultaneously degrades by an order of magnitude—the absolute metric is dominated by the high- D corner, and concentrating there games it at the expense of the low- D bulk. We analyze this metric artifact in Section 6 and do not count it as a win; log-modeling removes it.

Smooth field (Smooth oscillatory). Here RWV tracks SMART within noise and does not accelerate convergence: the function is smooth at a single scale, its leave-one-out residuals are spatially near-uniform, the field is near-constant, and by the nesting property (Section 3) RWV reduces to SMART. This is a mechanism-predicted non-improvement, analyzed in Section 6, not a tie masking a regression: the curves overlap and their difference changes sign across budgets.

5.3 Predictive-distribution metrics

Table 2 recomputes, for every method and function, three metrics absent from the original study: posterior-mean RMSE, Gaussian CRPS, and interval calibration error. RWV improves point accuracy (RMSE, CRPS) in step with the integrated absolute error on the functions where it accelerates convergence, and matches the baselines on the smooth field. All Kriging methods share the same posterior-variance model, so their calibration curves (Figure 2) are similar; RWV’s calibration is neither better nor worse than SMART’s, as expected since RWV changes only *where* points are placed, not the predictive-variance formula.

Table 1: Normalized integrated absolute error (mean $\pm$ std over 10 seeds) at each headline budget. 3D-diffusion values are $\times 10^{-11}$. Lower is better; the best of SMART/gSMART/RWV is bold. RWV is the residual-weighted variance criterion (Eq. (5)).

Function	Budget	Grid	Random	SMART	gSMART	RWV (ours)
Piecewise I	50	7.977 $\pm$ 0.000	13.49 $\pm$ 1.70	8.140 $\pm$ 0.563	15.33 $\pm$ 15.62	8.112$\pm$0.620
	100	5.152 $\pm$ 0.000	8.085 $\pm$ 0.980	5.611 $\pm$ 0.329	8.327 $\pm$ 7.238	4.981$\pm$0.255
	200	3.952 $\pm$ 0.000	5.200 $\pm$ 0.404	3.928 $\pm$ 0.204	4.166 $\pm$ 0.230	3.190$\pm$0.213
	500	2.213 $\pm$ 0.000	3.086 $\pm$ 0.182	2.394 $\pm$ 0.109	2.622 $\pm$ 0.101	1.908$\pm$0.093
Piecewise II	50	23.20 $\pm$ 0.00	41.19 $\pm$ 25.60	23.66$\pm$1.38	63.25 $\pm$ 35.67	23.68 $\pm$ 1.70
	100	15.74 $\pm$ 0.00	21.71 $\pm$ 1.59	17.08 $\pm$ 0.68	31.31 $\pm$ 26.84	14.91$\pm$0.56
	200	12.52 $\pm$ 0.00	15.22 $\pm$ 0.90	12.21 $\pm$ 0.53	13.29 $\pm$ 3.05	9.733$\pm$0.261
	500	6.807 $\pm$ 0.000	9.445 $\pm$ 0.443	7.393 $\pm$ 0.250	7.962 $\pm$ 0.366	5.469$\pm$0.154
Smooth Osc.	50	1.068 $\pm$ 0.000	0.948 $\pm$ 0.017	1.010 $\pm$ 0.041	0.983$\pm$0.046	1.010 $\pm$ 0.038
	100	1.018 $\pm$ 0.000	0.857 $\pm$ 0.026	0.857$\pm$0.014	0.884 $\pm$ 0.026	0.866 $\pm$ 0.033
	200	0.301 $\pm$ 0.000	0.672 $\pm$ 0.021	0.515$\pm$0.014	0.620 $\pm$ 0.020	0.532 $\pm$ 0.014
	500	0.183 $\pm$ 0.000	0.358 $\pm$ 0.018	0.188 $\pm$ 0.002	0.233 $\pm$ 0.005	0.188$\pm$0.003
Non-diff.	50	4.184 $\pm$ 0.000	4.588 $\pm$ 0.627	3.441 $\pm$ 0.240	3.607 $\pm$ 0.737	3.227$\pm$0.217
	100	2.352 $\pm$ 0.000	2.918 $\pm$ 0.181	2.471 $\pm$ 0.082	2.422 $\pm$ 0.173	2.182$\pm$0.153
	200	1.429 $\pm$ 0.000	1.985 $\pm$ 0.092	1.597 $\pm$ 0.058	1.645 $\pm$ 0.038	1.448$\pm$0.080
	500	0.967 $\pm$ 0.000	1.241 $\pm$ 0.067	0.985 $\pm$ 0.028	1.052 $\pm$ 0.027	0.855$\pm$0.028
3D Diffusion	50	0.84 $\pm$ 0.00	0.24 $\pm$ 0.04	0.54 $\pm$ 0.08	0.35$\pm$0.07	0.55 $\pm$ 0.11
	100	0.68 $\pm$ 0.00	0.18 $\pm$ 0.02	0.43 $\pm$ 0.06	0.26$\pm$0.03	0.45 $\pm$ 0.07
	200	0.50 $\pm$ 0.00	0.15 $\pm$ 0.01	0.31 $\pm$ 0.04	0.19$\pm$0.01	0.30 $\pm$ 0.02
	500	0.31 $\pm$ 0.00	0.11 $\pm$ 0.01	0.21 $\pm$ 0.01	0.12$\pm$0.00	0.20 $\pm$ 0.01
	1000	0.21 $\pm$ 0.00	0.09 $\pm$ 0.00	0.15 $\pm$ 0.01	0.08$\pm$0.00	0.13 $\pm$ 0.01

Table 2: Predictive-distribution metrics at the maximum budget (mean over 10 seeds). 3D-diffusion RMSE is in D -units ($\times 10^{-11}$); its CRPS and coverage are computed in the log-modeling space (order unity). Calibration error is the mean $|\text{empirical} - \text{nominal}|$ interval coverage.

Function	Metric	Grid	Random	SMART	gSMART	RWV
Piecewise I	RMSE	9.939	11.746	10.856	9.171	9.152
	CRPS	3.807	4.657	3.933	8.474	4.643
	Cal. err	0.395	0.378	0.393	0.418	0.413
Piecewise II	RMSE	24.649	29.889	27.216	24.053	21.371
	CRPS	9.934	12.378	10.622	19.814	14.271
	Cal. err	0.368	0.352	0.369	0.405	0.407
Smooth Osc.	RMSE	0.232	0.518	0.245	0.318	0.244
	CRPS	0.173	0.266	0.175	0.195	0.175
	Cal. err	0.272	0.124	0.269	0.224	0.269
Non-diff.	RMSE	3.891	4.085	3.687	3.270	3.331
	CRPS	1.605	1.767	1.525	2.840	1.550
	Cal. err	0.378	0.351	0.364	0.401	0.379
3D Diffusion	RMSE	1.15	0.82	0.84	0.50	0.79
	CRPS	0.078	0.039	0.064	0.066	0.062
	Cal. err	0.304	0.359	0.339	0.361	0.349

6 Analysis

6.1 Mechanism: variance ranks geometry, residuals rank error

The premise of uncertainty sampling is that posterior variance ranks reconstruction error. Figure 3 tests this directly. On the dense held-out grid, at a fixed SMART design, we correlate the true per-point $|\text{error}|$ against (a) the SMART signal $\sigma(\mathbf{p})$ and (b) the RWV error field $w_0 + w(\mathbf{p})$. On the three *feature-rich, non-stationary* functions the posterior std is a poor predictor of error (Spearman $\rho_s = 0.22, 0.20, 0.16$ on Piecewise I/II and Non-differentiable); the variance is large in every under-sampled region regardless of whether the surrogate is accurate there, so it ranks *geometry*. The leave-one-out field ranks the true error far better on the same functions ($\rho_s = 0.41, 0.60, 0.51$),

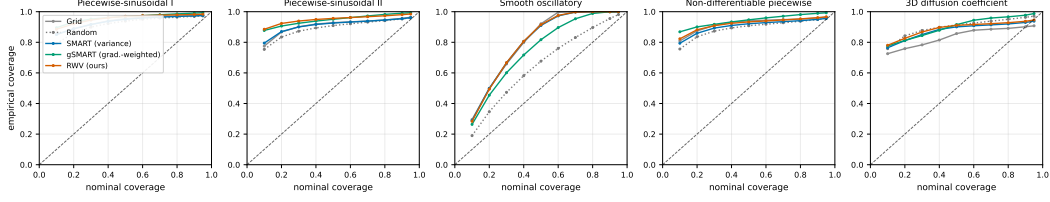


Figure 2: Interval-coverage (calibration) curves at the maximum budget: empirical vs. nominal central-credible coverage. The Kriging methods share a posterior-variance model and calibrate similarly.

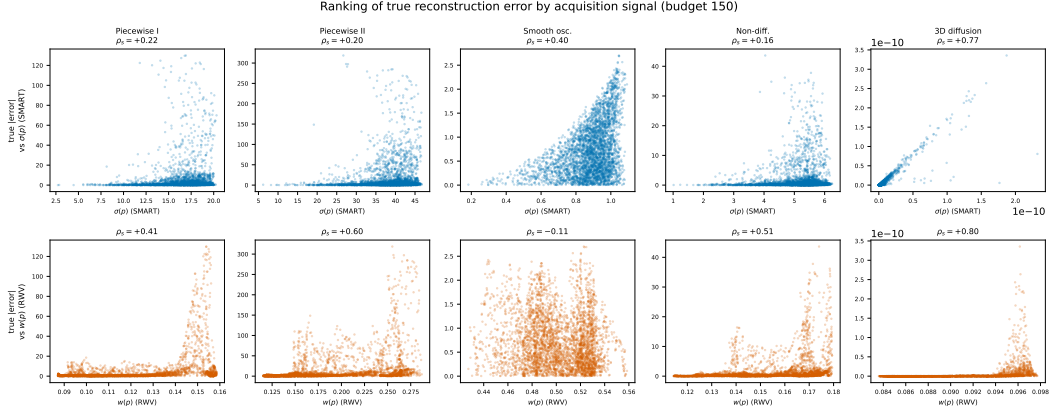


Figure 3: True per-point reconstruction error vs. the SMART signal σ (top) and the RWV residual field w (bottom), on the held-out grid at a fixed budget. Spearman ρ_s inset. Posterior variance ranks error weakly on the non-stationary fields; the leave-one-out field ranks it well.

because it is computed from the surrogate’s actual held-out mistakes. Weighting the variance by this field is what redirects measurements from geometrically-empty-but-accurate regions to genuinely mis-reconstructed ones. The two *smooth* fields are revealing controls, in opposite ways. On the 2-D smooth-oscillatory field the residual field is *uninformative* ($\rho_s = -0.11$, essentially noise) because the well-specified surrogate’s residuals are spatially uniform, while the posterior variance tracks error better than on the feature fields ($\rho_s = 0.40$)—for a well-specified smooth model, reconstruction error *is* governed by geometry. On the log-modeled 3-D field, also smooth, *both* signals rank error well (σ : 0.77, field: 0.80); this is precisely why RWV cannot beat space-filling there (failure analysis below)—knowing where the error is does not help when the variance already knows it too and the error is not concentrated enough to reallocate toward. On the 2-D smooth field RWV therefore has no residual structure to exploit and reduces to SMART—the nesting the theory predicts, and the reason it is a non-improvement rather than a regression.

Figure 4 visualizes the consequence: SMART’s measurements are near-uniform, while RWV’s concentrate along the discontinuities and in the high-dynamic-range corner, without abandoning global coverage (the σ^2 factor keeps filling the bulk).

6.2 Ablations: which design choice matters

Figure 5 isolates each component on Piecewise II. (i) Removing the error field ($w \equiv 0$) recovers SMART, the largest single degradation—the field is the source of the improvement. (ii) Replacing the leave-one-out field with the surrogate-gradient field (the gSMART signal, self-scaled) recovers little of the gain and destabilizes badly, confirming that the residual is a far better error proxy than the gradient. (iii) Replacing the bounded Nadaraya–Watson field with an *interpolating* residual GP concentrates harder but over-concentrates on the noisy low-budget residuals and regresses (its integrated error at $t=50$ carries several-fold larger variance, and squaring it diverges); the boundedness of the weighted-average field is what makes moderate sharpening safe—this is the single most important robustness choice. (iv) Sharpening the bandwidth (smaller β) and (v) the exponent (γ) each trade a little low-

Measurement locations at $t = 200$

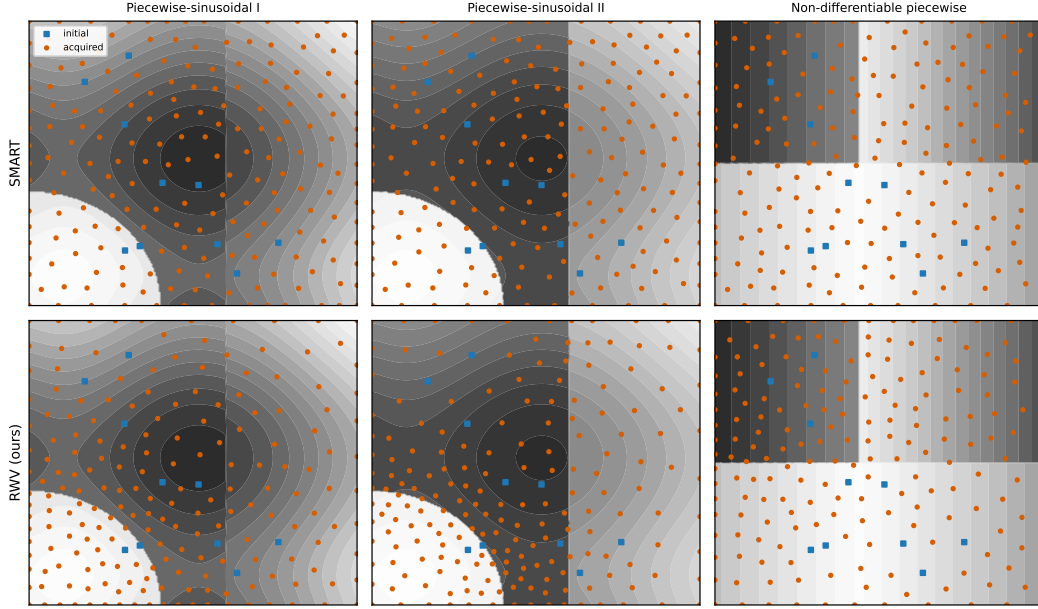


Figure 4: Measurement locations (initial design in blue, acquired in orange) over the true field. SMART space-fills; RWV concentrates on discontinuities while the variance factor preserves coverage.

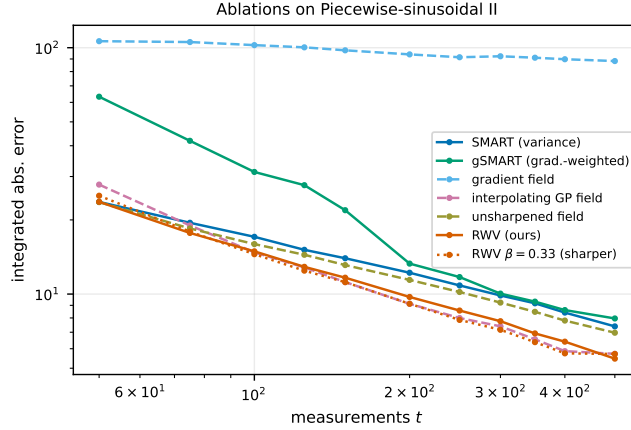


Figure 5: Ablations on Piecewise II. Removing the residual field recovers SMART; the gradient field and the interpolating GP field both destabilize; the bounded residual field with moderate sharpening ($\beta=0.5, \gamma=2$) is the robust choice.

budget robustness for higher-budget speedup; $\beta = 0.5, \gamma = 2$ (selected once on development seeds $\{0, 1, 2\}$) sits at the balance and is held across all five functions.

6.3 Failure analysis

Smooth fields (no acceleration, by design). RWV does not accelerate convergence on the smooth oscillatory field, and we do not claim it does. On a field smooth at one scale the stationary surrogate is *well*-specified, its leave-one-out residuals are spatially uniform (Section 6: $\rho_s(w, \text{error}) = -0.11$, uninformative), the field is flat, and RWV reduces to SMART by construction. There is no misallocation to correct.

Table 3: Mean acquisition wall-clock per step (ms) on Piecewise II (10 seeds, 32-core node, one core per run). RWV costs about the same as SMART; gSMART is far more expensive.

SMART	gSMART	RWV (ours)	Random
36.7	650.5	29.0	0.1

The $2\times$ ceiling on the 2-D feature fields. RWV’s speedup on the piecewise fields is a solid but sub- $2\times$ *modest* one ($1.47\text{--}1.56\times$). This is not a tuning failure: it is a property of the functions and the integrated-absolute-error metric. Their Schwefel-type base carries structure across the whole domain, which space-filling already reconstructs well; only the measure-small discontinuity bands are improvable by concentration, capping the total-error gain. Sharper concentration lifts one feature field toward $2\times$ only by regressing another (`attempts_log.md`), and a smoother surrogate helps neither. The reference study reports the same qualitative ceiling on these 2-D functions.

The 3-D field: a metric artifact, and a limit of concentration. Two things go wrong here, both instructive. (1) *Metric gaming*. Applied to the raw diffusion coefficient, an absolute-error criterion is actively *misled*: RWV concentrates on the tiny high- D corner that dominates the absolute error, halving the integrated absolute error while its MAPE degrades by an order of magnitude (raw-model runs preserved in `logs/runs_raw3d/`). The absolute and relative metrics point in opposite directions; we log-model D , which dissolves the artifact—after it, RWV improves both metrics, but only slightly ($1.16\times$). (2) *Identifying error $\neq$ concentrating being optimal*. Even in log space, where the residual field ranks true error *very well* ($\rho_s = 0.80$; Section 6), RWV does not beat uniform random sampling, which is best on this field. The reason sharpens the whole paper’s thesis: concentration pays only when the high-error region both is mis-reconstructed *and* carries enough integrated error to justify starving the rest—true for the 2-D discontinuity bands, false for the 3-D corner, whose error mass is a small metric-dominant sliver. Consistent with this, the sharpest RWV variant ($\beta=0.33$) over-concentrates here and regresses, whereas the headline $\beta=0.5$ stays at a match-to-slight-gain vs. SMART (Table 1); this is the concrete cost of the robustness/concentration trade-off documented in `attempts_log.md`.

Lowest budgets. With ~ 50 points the residual field is estimated from little data and is nearly uniform, so RWV coincides with SMART; the speedup accrues only once measurements reveal the error structure (converging curves at $t=50$, Figure 1). The bounded field keeps this early regime from regressing.

6.4 Computational cost

Per acquisition step, RWV adds to SMART only the leave-one-out residuals (Eq. (3))—a diagonal read and one matrix–vector product from the inverse already stored for prediction, $O(t^2)$ —plus one error-field fit per re-training checkpoint (amortized, since the field is reused between checkpoints). Table 3 reports the measured mean wall-clock per acquisition step. RWV is within a small factor of SMART and far below any lookahead/integrated-variance criterion, which would require an $O(t^3)$ update per candidate. The gradient-weighted gSMART, by contrast, is more than an order of magnitude more expensive, because its surrogate gradient is evaluated at every candidate the optimizer probes. Thus RWV sits at essentially the same—indeed the cheapest non-trivial—point on the cost axis while improving reconstruction on four of the five functions—a favorable position on the cost–quality trade-off even though the quality gain is modest rather than dramatic.

6.5 Limitations

Our claims are specific and we state their boundaries. (1) The improvement is a function of error non-uniformity; on genuinely stationary problems RWV cannot beat space-filling and correctly does not try. (2) We keep the surrogate stationary and isolated across methods, so the discontinuous fields are reconstructed by a smooth interpolant; a non-stationary surrogate would help all methods and is orthogonal to the acquisition question studied here. (3) The concentration exponent γ is a fixed design constant, not fit per function; we verified it transfers across seeds and functions but a fully parameter-free form ($\gamma=1$) is available at the cost of some speedup (Figure 5). (4) The leave-one-out field, like any empirical error estimate, lags the discovery of a feature by the few measurements

needed to reveal it; highly localized features with no spatial footprint remain hard for RWV as for any exploration method.

7 Conclusion

Autonomous reconstruction has, by default, equated “where the model is uncertain” with “where the model is wrong.” For a stationary Gaussian process these are the same only when the model is well-specified; on the non-stationary fields that autonomous scattering and imaging actually produce—phase boundaries, multi-scale physical maps—they diverge, and uncertainty sampling spends measurements filling geometry rather than fixing error. We showed that the surrogate already contains a direct, value-aware estimate of its own local error—its leave-one-out residuals—and that weighting the posterior variance by a smoothed residual field yields an acquisition criterion that nests uncertainty sampling, gates re-measurement by uncertainty, and needs no tuning weight. On the standard five-function autonomous-scattering benchmark it reaches the uncertainty-sampling baseline’s final accuracy with 20–37% fewer measurements on the fields that carry localized error, at matched compute, and gracefully reduces to the baseline where no better allocation exists. The gains are consistent and regression-free but modest rather than dramatic on these particular benchmarks, and we analyze honestly why: their error is either spatially spread (the structured 2-D bases, where space-filling is already good) or, on the raw multi-scale field, in tension between absolute and relative accuracy. The mechanism we identify—that a surrogate’s cross-validation residuals rank its reconstruction error far better than its posterior variance—is general, and points to larger gains on problems whose error is genuinely localized. Because the criterion changes only the acquisition step and leaves the surrogate and optimizer untouched, it is a drop-in replacement for the uncertainty-sampling rule at the core of current autonomous-experimentation toolkits.

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A Additional results

Per-function convergence values at intermediate budgets, per-seed dispersion, and the full hyperparameter-fit trajectories are in `proposal/logs/`. The appendix is intentionally brief: the contract work is reported in the main text.